On Device Personal Health Assistant

A neurosymbolic split for yoga coaching: measure the pose, learn the score from the measurement, generate only the cue.

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A vision-language model asked to name a yoga pose invents both the name and the score. Recover the name from 33 landmarks, learn the score on-device from those same landmarks, and let the model write only the cue. Nothing past the skeleton sees a pixel.

CONTRIBUTIONS

1. A 12-D joint-angle descriptor, scale-invariant, so body size never enters the representation.
2. Deterministic pose identity with explicit abstention; the label stops being generated text.
3. A symbolic bottleneck: every learned stage downstream sees angles, never pixels.

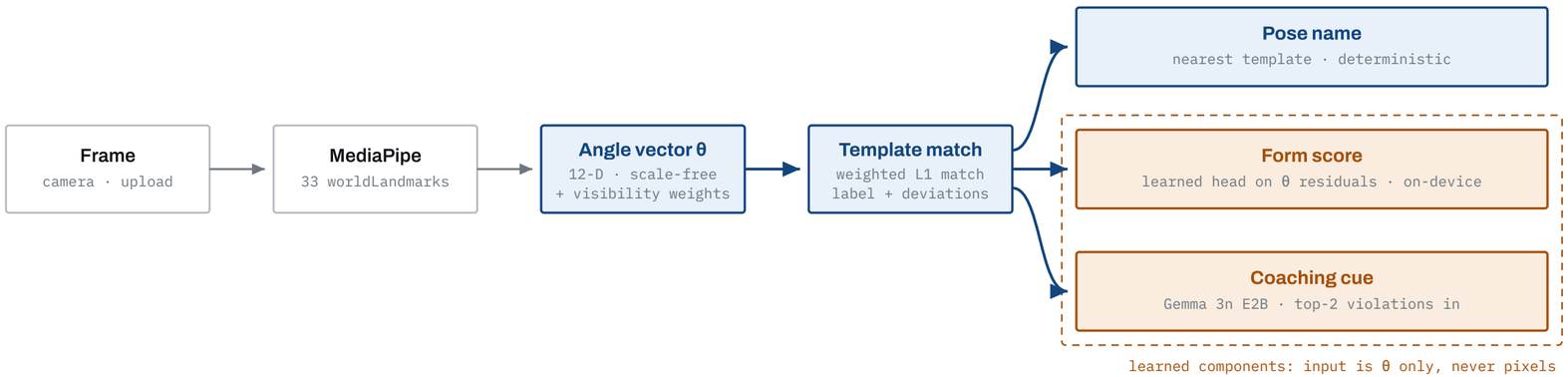


Figure 1. Landmarks are never rendered back into pixels, so they survive as symbols. Identity is retrieval and is settled by geometry; quality is a judgement and stays learned, but learned *from the angle deviations*. Blue is symbolic, amber is learned; the dashed boundary marks everything whose input is θ alone.

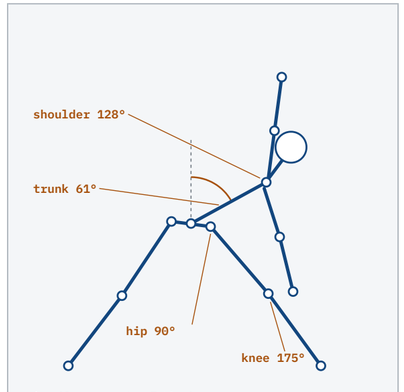


Figure 2. The descriptor on a Trikonasana variant. Gemma labelled this frame “Warrior III”, a one-legged balance it does not resemble, and scored it 8/10. Every angle here contradicts that label, and none of them reached the model.

1 A 12-D descriptor

Interior angles are scale-invariant: stature and limb length drop out with no normalisation. Metric `worldLandmarks` [1] remove most camera-azimuth dependence, a confound that otherwise looks like individual variation.

COMPONENT	LANDMARKS
elbow L / R	`11·13·15 / 12·14·16`
shoulder L / R	`13·11·23 / 14·12·24`
hip L / R	`11·23·25 / 12·24·26`
knee L / R	`23·25·27 / 24·26·28`
trunk tilt	`hip→shldr vs. ĝ`
stance ratio	$\|a_L - a_R\|$ / shldr

Table 1. Descriptor components, with MediaPipe landmark indices.

2 Matching

$$d(q,t) = \Sigma_i\, w_i \cdot |q_i - t_i| \;/\; \Sigma_i\, w_i$$

w_i = per-pose importance × landmark visibility

The front knee defines Warrior II; the elbow is noise. Visibility inside w_i means an occluded joint **stops voting rather than voting wrongly**. The signed residuals feed the scorer.

3 Abstention

ABSOLUTE Distance above 25–30° yields `UNRECOGNIZED`.

MARGIN A narrow first-to-second gap means ambiguity; hedge, don’t assert.

TEMPORAL Classify during holds only. Transitions carry no descriptor.

Abstention rate per subgroup is itself a fairness measurement.

4 What the learned stages see

```
pose   Virabhadrasana II (right)  matched, d = 14°
score  6.4 / 10                   learned head

violations   ranked by w_i·|q_i − t_i|
 KNEE_FRONT  118°  tgt  90 ± 15  +28
 TRUNK_TILT   19°  tgt   0 ± 12   +7
 KNEE_BACK   173°  tgt 175 ± 10   ok
 SHOULDER_L   88°  tgt  90 ± 15   ok
```

The scorer reads the residual vector; the language model reads the top two rows. Neither receives an image, coordinates, or a label to guess.

And what comes back

> *“Sink your hips until the front thigh is parallel. You’re riding about thirty degrees high, and it’s pulling your chest forward.”*

5 Fairness by construction

Every learned stage sees only symbolic facts. None can condition on skin tone, body size, gender presentation, or clothing, because *those features are not in the input*.

An end-to-end VLM can condition on all of them and cannot report whether it did. Point targets encode a physique: long femurs make an upright Utkatasana impossible. Use **tolerance bands widened by measurement uncertainty**.

`L ↔ R`	Asymmetry compares a person to themselves; anthropometry cancels exactly.
$t_0 \leftrightarrow t_n$	Within-session drift measures fatigue against the same body.
`COM ⊂ BOS`	Mass over base of support is physics, not aesthetics.

6 Limitations

Templates are hand-authored, encoding one reading of alignment; lineage should be selectable. The scorer needs labelled form ratings, which carry the same normative bias. Upstream landmark error is not expected to be uniform across skin tone [5], body size or clothing, and per-subgroup rates are not yet reported.

[1] Bazarevsky et al. BlazePose. CVPRW on CV for AR/VR, 2020. arXiv:2006.10204 [2] Lugaresi et al. MediaPipe. arXiv:1906.08172, 2019 [3] Gemma Team, Google DeepMind. Gemma 3n, 2025 [4] Verma, Kumawat, Nakashima, Raman. Yoga-82. CVPR Workshops, 2020 [5] Monk. The Monk Skin Tone Scale. SocArXiv, 2023. doi:10.31235/osf.io/pdf4c